

Meridian University

Department of Computer Science

CS 301: Algorithms & Data Structures

Final Examination · Fall Term 2026

✓ SOLUTION KEY

Instructor: Dr. Elena Whitfield **Duration:** 120 minutes **Total Points:** 100

Term: Fall 2026 **Materials:** One handwritten note card **Date:** December 11, 2026

 Instructions

- Write all answers in the space provided; overflow work may continue on the back of the page if clearly labeled.
- Show complete work for Parts III and IV — unsupported final answers receive at most half credit.
- Calculators are permitted; network-connected devices are **not**.
- Once the proctor calls time, put down all writing instruments immediately.
- Academic integrity: this exam is governed by the Meridian University Honor Code (Section 4.2). Submitting this exam constitutes an affirmation that you neither gave nor received unauthorized assistance.

Part	Topic	Points	Score
I	Multiple Choice	10	—
II	Short Answer	20	—
III	Problem Solving & Proofs	30	—
IV	Case Study: Data-Center Routing	40	—
Total		100	—

Part I

Multiple Choice

10 points

Question 1. What is the worst-case time complexity of binary search on a sorted array of n elements? [2 points]

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(\log n)$